Graph-SND: Sparse Aggregation for Behavioral Diversity in Multi-Agent Reinforcement Learning

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Abstract

System Neural Diversity (SND) measures behavioral heterogeneity in multi-agent reinforcement learning by averaging pairwise distances over all $\binom{n}{2}$ agent pairs, making each call quadratic in team size. We introduce Graph-SND, which replaces this complete-graph average with a weighted average over the edges of an arbitrary graph G. Three regimes follow: $G = K_n$ recovers SND exactly; a fixed sparse G defines a localized diversity measure at $O(|E|)$ cost; and random edge samples yield an unbiased Horvitz–Thompson estimator and a normalized sample mean with $O(1/\sqrt{m})$ concentration in the sampled edge count m. For fixed sparse graphs we prove forwarding-index distortion bounds for expanders and a spectral refinement under low-rank distance structure; for random d-regular graphs we prove an unconditional probabilistic $\widetilde{\mathcal{O}}(D_{\max}/\sqrt{n})$ bound. On VMAS we verify recovery, unbiasedness, concentration, and wall-clock scaling, with a PettingZoo TVD panel checking non-Gaussian transfer. In a 500-iteration $n = 100$ PPO run, Bernoulli-0.1 Graph-SND tracks full SND while reducing per-call metric time by about $10\times$, and frozen-policy GPU timing up to $n = 500$ follows the predicted $\binom{n}{2}/|E|$ speedup. Random d-regular expanders empirically achieve $\mathrm{SND}^{\mathrm{u}}_G/\mathrm{SND} \in [0.9987, 1.0013]$ at $\Theta(n \log n)$ edges. In DiCo diversity control at $n = 50$, Bernoulli-0.1 Graph-SND preserves set-point tracking with paired reward differences indistinguishable from zero across nine matched cells while cutting per-call metric cost by $\sim 9.5\times$. Together, these results show that the SND aggregation bottleneck can be removed without changing the metric's semantics, yielding a drop-in sparse alternative that scales beyond complete-graph SND and supports both passive measurement and closed-loop diversity control.

1 Introduction

Behavioral diversity helps multi-agent reinforcement learning (MARL) systems specialize, remain robust, and expose capabilities not visible in reward alone [Bettini et al., 2023, 2025]. System Neural Diversity (SND) [Bettini et al., 2025] measures this heterogeneity by averaging pairwise Wasserstein distances between all agents' action distributions and enables diversity control through DiCo [Bettini et al., 2024a]. Its bottleneck is aggregation: every call uses all $\binom{n}{2}$ agent pairs, and Bettini et al. [2025, §6] identify subgroup-to-system aggregation as open.

We introduce Graph-SND, which replaces SND's complete-graph average by a weighted average over the edges of a graph G. The complete graph recovers SND exactly. A fixed graph, such as a communication or nearest-neighbor graph, defines a localized diversity measure over task-relevant interactions. A random graph defines sparse estimators of full SND, including an unconditionally unbiased Horvitz–Thompson estimator and a normalized sample mean with finite-population con-

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centration. This isolates the aggregation bottleneck while leaving the policy architecture, training algorithm, and pairwise distance unchanged.

Contributions. (i) We define Graph-SND (Definition 1), a sparse aggregation layer that recovers SND on K_n and costs $O(|E|)$ on a precomputed graph. (ii) We prove recovery, non-negativity, graph-automorphism invariance, Horvitz–Thompson unbiasedness, conditional concentration, forwarding-index distortion bounds [Heydemann et al., 1989, Leighton and Rao, 1999], a spectral nuclear-norm refinement, and an unconditional probabilistic $\widetilde{\mathcal{O}}(D_{\max}/\sqrt{n})$ bound for random d-regular graphs. (iii) VMAS experiments [Bettini et al., 2022] verify recovery, concentration, and wall-clock scaling through $n = 500$, while PettingZoo provides a discrete-action TVD transfer check; a 500-iteration $n = 100$ PPO run shows Bernoulli-0.1 Graph-SND tracking full SND at about $10\times$ lower metric cost. (iv) Random d-regular expanders achieve $\mathrm{SND}^{\mathrm{u}}_G/\mathrm{SND} \in [0.9987, 1.0013]$ at $\Theta(n \log n)$ edges, and DiCo at $n = 50$ shows Bernoulli-0.1 Graph-SND preserves set-point tracking and yields paired reward differences indistinguishable from zero across nine matched cells while reducing per-call metric cost by $\sim 9.5\times$.

2 Related Work

Diversity measures in multi-agent RL. Prior metrics compare policy distributions [McKee et al., 2022, Yu et al., 2021, Hu et al., 2022], occupancy measures [Liu et al., 2021], deterministic action diversity [Parker-Holder et al., 2020], or trajectory distributions [Lupu et al., 2021]. SND [Bettini et al., 2025] is closest: it is a metric on concurrently acting Gaussian policies and is used by DiCo [Bettini et al., 2024a], measurement studies [Bettini et al., 2024b, Tessera et al., 2024], and diversity-benefit analysis [Amir et al., 2025]. These works treat the all-pairs average as fixed. Graph-SND changes only this aggregation layer.

Heterogeneity in robotics. Robotics heterogeneity is often structural rather than behavioral: species-trait matrices [Prorok et al., 2017], performance proxies [Li et al., 2004], and fixed-class counts [Twu et al., 2014]. Hierarchic Social Entropy [Balch, 2000] is closer, but Bettini et al. [2025] show it does not capture redundancy in the same way as SND. Graph-SND inherits SND's complete-graph behavior and adds a structured sparse alternative.

Graphs in multi-agent learning, and subgroup partitioning. Graphs in MARL usually appear inside policies as communication or message-passing structure [Sukhbaatar et al., 2016, Blumenkamp and Prorok, 2021, Bettini et al., 2023] or as role/credit inductive bias [Wang et al., 2020, 2021]. Our graph is external: it selects which pairwise behavioral distances enter the scalar diversity aggregate. Subgroup partitions proposed by Bettini et al. [2025, §6] are a special case (block-diagonal unions of cliques), while k-NN, communication, Bernoulli, and expander graphs are not representable as simple partitions.

Sampling-based approximations, and follow-up work on SND. The random-G view uses Horvitz–Thompson estimation [Horvitz and Thompson, 1952], Hoeffding's finite-population concentration [Hoeffding, 1948, 1963], and Serfling-style refinements [Serfling, 1974, Bardenet and Maillard, 2015]. The fixed-expander view connects MARL diversity aggregation to edge forwarding index and multicommodity-flow structure [Chung et al., 1987, Heydemann et al., 1989, Leighton and Rao, 1999], which prior pairwise diversity metrics do not exploit.

3 Background

3.1 Partially Observable Markov Games

We work in cooperative Partially Observable Markov Games [Kaelbling et al., 1998] with agents $\mathcal{A} = \{1, \ldots, n\}$, private observations, shared dynamics, and heterogeneous stochastic policies π_{θ_i}. Agents share observation and action spaces so their action distributions can be compared at the same observation; heterogeneous spaces would require an alignment map and are outside this paper's scope. The diversity metrics are computed on rollout data from independently parameterized policies, as in Bettini et al. [2023, 2025].

3.2 System Neural Diversity

Following Bettini et al. [2025], let $\mathcal{A} = \{1, \ldots, n\}$ be the agent set and let each agent $i \in \mathcal{A}$ have a Gaussian policy $\pi_{\theta_i}(o) = \mathcal{N}\big(\mu_{\theta_i}(o), \Sigma_{\theta_i}(o)\big)$. Let $\mathcal{B} = \{o^t\}_{t=1}^T$ be a collection of joint observation vectors sampled from policy rollouts, where each $o^t = (o_1^t, \ldots, o_n^t) \in \mathcal{O}^n$. The Monte Carlo behavioral distance between agents i and j is

$$d(i,j) \;=\; \frac{1}{|\mathcal{B}|\,|\mathcal{A}|} \sum_{o^t \in \mathcal{B}} \sum_{k \in \mathcal{A}} W_2\big(\pi_{\theta_i}(o_k^t),\, \pi_{\theta_j}(o_k^t)\big), \tag{1}$$

where W_2 admits a closed form on multivariate Gaussians [Bettini et al., 2025]. The global System Neural Diversity is

$$\mathrm{SND}(\mathbf{D}) \;=\; \binom{n}{2}^{-1} \sum_{i<j} d(i,j), \tag{2}$$

the mean of pairwise behavioral distances over unique agent pairs. We work with (1) directly rather than its integral form, avoiding specification of a measure on $\mathcal{O}$; all our results are stated at the level of the sampled estimator. Throughout, we treat the values $\{d(i,j)\}_{\{i,j\} \in \mathcal{P}}$ as fixed (deterministic in $\mathbf{D}$), so that all stochasticity in the estimators below arises solely from the random graph.

4 Graph-SND

Full SND weights every pair equally. This is both computationally prohibitive (quadratic in n) and, in many settings, semantically imprecise: agents that do not interact or operate on disjoint subtasks contribute to the score on equal footing with coupled pairs. We introduce a graph-structured generalization that supports two distinct use cases: a localized diversity measure when G encodes interaction structure, and sampling-based estimators of SND when edges of G are drawn from the pair set.

4.1 Definition

Let $\mathcal{P} := \binom{\mathcal{A}}{2}$ be the set of unordered agent pairs, and let $G = (\mathcal{A}, E, w)$ be an undirected graph on the agent set with $E \subseteq \mathcal{P}$ and non-negative edge weights $w : E \to \mathbb{R}_{\geq 0}$. Write $W(G) = \sum_{\{i,j\} \in E} w_{ij}$ and assume $W(G) > 0$. The graph may be fixed (e.g., communication topology, k-NN on observations, ε-ball) or sampled from pairs (Section 5.2).

Definition 1 (Graph-SND)**.** The Graph-SND of policies $\{\pi_{\theta_i}\}_{i \in \mathcal{A}}$ on G is

$$\mathrm{SND}_G(\mathbf{D}, G) \;=\; \frac{1}{W(G)} \sum_{\{i,j\} \in E} w_{ij}\, d(i,j), \tag{3}$$

with $d(i,j)$ given by (1).

When $G = K_n$ with unit weights, $W(G) = \binom{n}{2}$ and $\mathrm{SND}_G(\mathbf{D}, K_n) = \mathrm{SND}(\mathbf{D})$ identically.

4.2 Canonical graph families

Three choices of G recover interpretable special cases.

- **Complete graph**, $G = K_n$, $w \equiv 1$: recovers SND.
- k**-nearest-neighbor graph** in observation space: $|E| = O(nk)$, yielding a $\Theta(n/k)$ reduction in the number of pairwise distance evaluations relative to full SND.
- **Communication graph** inherited from the policy's message-passing topology: aligns measured diversity with the interaction structure actually used by the agents.

In each case G may be fixed or state-dependent; the latter requires re-evaluating SND_G per rollout batch.

For any weighted graph $G' = (\mathcal{A}', E', w')$ with $W(G') = 0$, we define

$$\mathrm{SND}_G(\mathbf{D}, G') := 0$$

by convention.

5 Theoretical Properties

We state Graph-SND's core properties, then give a probabilistic interpretation as sampling-based estimators of SND with concentration guarantees. Proofs are in Appendix A.

5.1 Core properties

Proposition 2 (Recovery on the complete graph)**.** *If $G = K_n$ with unit edge weights, then* $\mathrm{SND}_G(\mathbf{D}, G) = \mathrm{SND}(\mathbf{D})$.

Proposition 3 (Non-negativity)**.** $\mathrm{SND}_G(\mathbf{D}, G) \geq 0$*, with equality if and only if* $d(i,j) = 0$ *for every edge* $\{i,j\} \in E$ *with* $w_{ij} > 0$.

Remark 4 (Semantic shift from SND). Proposition 3 is strictly weaker than the zero condition of full SND: SND_G can vanish even when the system contains behaviorally distinct agents, provided those agents are not adjacent in G. When G encodes task-relevant interaction structure, this is the intended semantics: Graph-SND measures *locally relevant* diversity. When full-system diversity is the target, the estimator interpretation of Section 5.2 is the appropriate one.

Proposition 5 (Graph-automorphism invariance)**.** *For any automorphism σ of (G, w),* $\mathrm{SND}_G(\sigma \cdot \mathbf{D}, G) = \mathrm{SND}_G(\mathbf{D}, G)$*, where* $(\sigma \cdot \mathbf{D})_{ij} = d(\sigma(i), \sigma(j))$.

Proposition 6 (Complexity)**.** *Evaluating* SND_G *on a precomputed graph G requires $|E|$ pairwise distance computations $d(\cdot,\cdot)$, versus $\binom{n}{2}$ for* SND. *For k-NN graphs with $|E| = O(nk)$, this yields a speedup factor of $\Theta(n/k)$ in the number of pairwise distance evaluations. This statement excludes the cost of constructing G; measured construction overheads for the graph builders used here appear in Appendix C (Table 6).*

5.2 Sampling estimators of SND

When the goal is to approximate full SND at reduced cost, we derive sampling-based estimators from the same graph formalism. Let G_p include each pair $\{i,j\} \in \mathcal{P}$ independently with probability $p \in (0,1]$. We distinguish two random-graph statistics. The Horvitz–Thompson statistic uses the population normalization $|\mathcal{P}|^{-1}$ and is unbiased even when the sampled edge set is empty (the empty sum contributes zero). The normalized Graph-SND statistic uses the realized sample size $m = |E(G_p)|$ and is the sample mean over sampled edges; it is defined conditional on $m \geq 1$ and is the quantity used in the concentration theorem below.

Proposition 7 (Unbiasedness)**.** *With weights $w_{ij} = 1/p$, the Horvitz–Thompson estimator*

$$\widehat{\mathrm{SND}}_{\mathrm{HT}}(G_p) \ := \ |\mathcal{P}|^{-1} \sum_{\{i,j\} \in E(G_p)} w_{ij}\, d(i,j)$$

satisfies $\mathbb{E}_{G_p}\big[\widehat{\mathrm{SND}}_{\mathrm{HT}}(G_p)\big] = \mathrm{SND}(\mathbf{D})$.

Remark 8 (Normalized Graph-SND is a sample mean). If the same $1/p$ weights are inserted into the normalized Graph-SND definition, $W(G_p) = |E(G_p)|/p$, so the weights cancel and SND_G reduces to the uniform sample mean:

$$\mathrm{SND}_G(\mathbf{D}, G_p) \ = \ \frac{p}{|E(G_p)|} \sum_{\{i,j\} \in E(G_p)} \frac{1}{p}\, d(i,j) \ = \ \frac{1}{|E(G_p)|} \sum_{\{i,j\} \in E(G_p)} d(i,j).$$

This *uniform-weight variant* differs from $\widehat{\mathrm{SND}}_{\mathrm{HT}}$ in its random normalization. It is used in Theorem 9 conditional on $|E(G_p)| \geq 1$. In the DiCo Bernoulli implementation, an empty draw falls back to the full pair set for that call so the control ratio remains well-defined.

Theorem 9 (Concentration)**.** *Assume $d(i,j) \in [0, D_{\max}]$ for all $\{i,j\} \in \mathcal{P}$. Let $m = |E(G_p)|$, and define the uniform-weight variant*

$$\mathrm{SND}^{\mathrm{u}}_G(\mathbf{D}, G_p) \ := \ m^{-1} \sum_{\{i,j\} \in E(G_p)} d(i,j).$$

Then for any $\delta \in (0,1)$, conditional on $|E(G_p)| = m \geq 1$, with probability at least $1 - \delta$ over the randomness of G_p, Hoeffding's sampling-without-replacement finite-population extension gives

$$|\mathrm{SND}^{\mathrm{u}}_G(\mathbf{D}, G_p) - \mathrm{SND}(\mathbf{D})| \ \leq \ D_{\max}\sqrt{\frac{\log(2/\delta)}{2m}}. \tag{4}$$

Remark 10 (Unconditional Bernoulli form). Let $\mu = p|\mathcal{P}|$ be the expected number of sampled edges in G_p. Combining Theorem 9 with the Chernoff bound $\mathbb{P}(|E(G_p)| < \mu/2) \leq e^{-\mu/8}$ gives, for any $\delta \in (0,1)$, with probability at least $1 - \delta - e^{-\mu/8}$,

$$|\mathrm{SND}^{\mathrm{u}}_G(\mathbf{D}, G_p) - \mathrm{SND}(\mathbf{D})| \;\leq\; D_{\max}\sqrt{\frac{\log(2/\delta)}{\mu}},$$

on the event $|E(G_p)| \geq 1$. Thus Bernoulli sampling has the same finite-population $O(1/\sqrt{p|\mathcal{P}|})$ rate up to constants whenever the expected edge count is not tiny.

Remark 11 (Sharper finite-population bounds). For sampling fractions $m/|\mathcal{P}|$ bounded away from zero, Serfling's inequality [Serfling, 1974] gives a bound of the form $D_{\max}\sqrt{(1-(m-1)/|\mathcal{P}|)\log(2/\delta)/(2m)}$, which is strictly sharper. See also Bardenet and Maillard [2015] for further refinements and an empirical-Bernstein variant.

Theorem 9 gives a conditional rate of $O(1/\sqrt{m})$ in the number of sampled edges. For target error ε, it suffices to sample $m = O\big(D^2_{\max}\,\varepsilon^{-2}\,\log(1/\delta)\big)$ edges, which for large n is vastly smaller than $\binom{n}{2}$.

For fixed sparse graphs, let $\pi(G)$ denote the edge forwarding index: the minimum over all pairwise path routings of the maximum number of routed paths using any edge [Chung et al., 1987, Heydemann et al., 1989].

Theorem 12 (Deterministic fixed-G distortion)**.** *Let $G = (\mathcal{A}, E)$ be connected with unit edge weights, and suppose $d : \mathcal{P} \to \mathbb{R}_{\geq 0}$ satisfies the triangle inequality. With* $\mathrm{SND}^{\mathrm{u}}_G(\mathbf{D}, G) := |E|^{-1}\sum_{\{i,j\}\in E} d(i,j)$,

$$\frac{|E|}{|\mathcal{P}|}\mathrm{SND}^{\mathrm{u}}_G(\mathbf{D}, G) \leq \mathrm{SND}(\mathbf{D}) \leq \frac{|E|\pi(G)}{|\mathcal{P}|}\mathrm{SND}^{\mathrm{u}}_G(\mathbf{D}, G). \tag{5}$$

Thus $|E|/|\mathcal{P}| \leq \mathrm{SND}/\mathrm{SND}^{\mathrm{u}}_G \leq |E|\pi(G)/|\mathcal{P}|$ whenever $\mathrm{SND}^{\mathrm{u}}_G > 0$, and both ratios are 1 on K_n.

Corollary 13 (Expander sparsification)**.** *For a d-regular spectral expander, $\pi(G) = \mathcal{O}(n\log n/d)$ [Leighton and Rao, 1999], so Theorem 12 gives $\mathcal{O}(\log n)$ worst-case relative distortion using only $|E| = \Theta(n\log n)$ pairwise distances when $d = \Theta(\log n)$.*

Remark 14 (Spectral sharpening). Proposition 17 (Appendix A) sharpens this fixed-graph guarantee under controlled normalized nuclear norm $\rho_*(\mathbf{D}) = \|\mathbf{D}\|_*/(n\,\mathrm{SND}(\mathbf{D}))$: for d-regular expanders, the relative error scales as $\mathcal{O}(\rho_*(\mathbf{D})/\sqrt{d})$, hence $\mathcal{O}(\rho_*/\sqrt{\log n})$ at $d = \Theta(\log n)$. This does not prove the empirical near-unit ratios, but it explains why low-effective-rank MARL distance matrices can be much easier than the worst case.

Remark 15 (Unconditional probabilistic bound). Proposition 19 (Appendix A) gives a fully unconditional concentration bound: for G drawn uniformly from simple d-regular graphs, with probability at least $1-\delta$, $|\mathrm{SND}^{\mathrm{u}}_G - \mathrm{SND}| \leq \widetilde{\mathcal{O}}(D_{\max}/\sqrt{n})$ at $d = \Theta(\log n)$, requiring only $D_{ij} \in [0, D_{\max}]$. This provides a distribution-free probabilistic complement to the worst-case $\mathcal{O}(\log n)$ ratio of Corollary 13, though sharper distribution-dependent analysis would be needed to fully explain the empirically observed near-unit ratios in Section 6.3.

Both Theorems 9 and 12 depend only on boundedness (and for Theorem 12, triangle inequality) of $d(i,j)$, not on the Gaussian/Wasserstein instantiation; discrete action spaces may substitute bounded metrics such as total variation or the square-root Jensen–Shannon metric.

6 Experiments

We evaluate Graph-SND as an aggregation layer, so the experiments target metric recovery, estimator scaling, deterministic sparse approximation, and downstream DiCo control rather than new task-level RL performance. Hyperparameters and rollout details are in Appendix B. Core checks on VMAS Multi-Agent Goal Navigation verify exact recovery on K_4, Horvitz–Thompson unbiasedness at $n = 8$, and zero concentration-bound violations across 12 cells with 2,000 finite-population draws each at $n = 16$ (Appendix C).

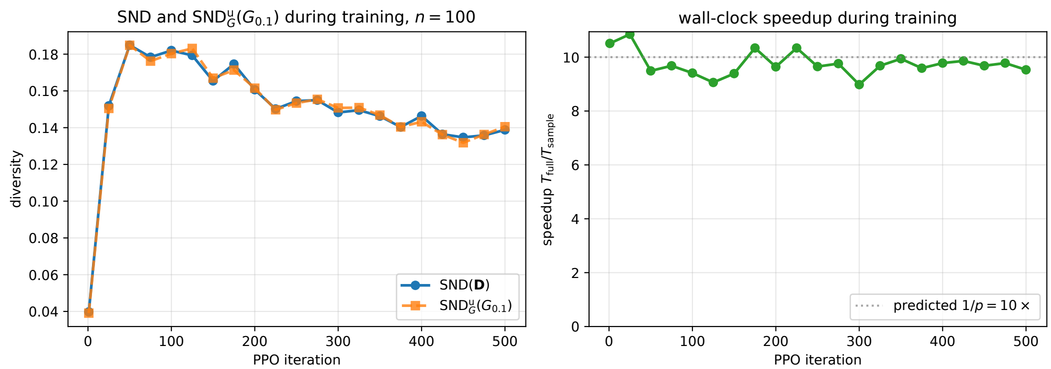

Figure 1: $n = 100$ PPO training run. Bernoulli-0.1 Graph-SND tracks full SND at every logged iteration while reducing metric time by about $10\times$, matching Proposition 6 under non-stationary training dynamics.

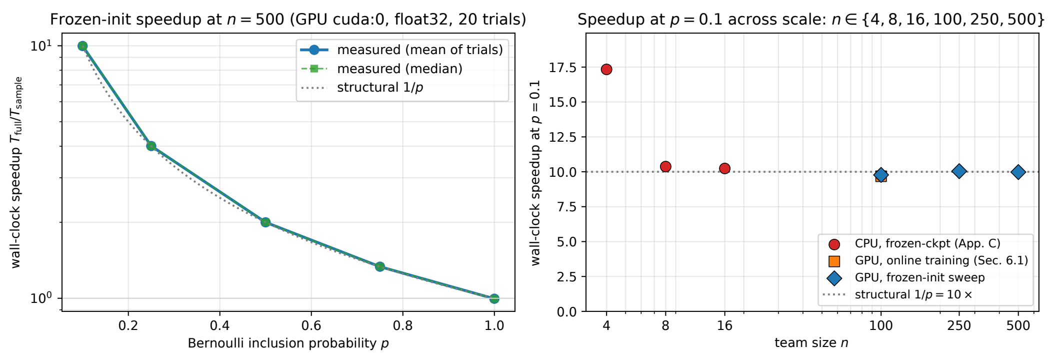

Figure 2: Wall-clock scaling of $\mathrm{SND}^{\mathrm{u}}_G(G_p)$ versus full SND across $n \in \{4, \ldots, 500\}$. Bernoulli-p sampling follows the structural $1/p$ speedup, and Bernoulli-0.1 remains near $10\times$ through $n = 500$; sampled timings include edge sampling and edge-transfer overhead.

6.1 Scaling to $n = 100$ under training dynamics

We train 100 independently parameterized Gaussian policies on VMAS Multi-Agent Goal Navigation for 500 PPO iterations. Every 25 iterations we log full SND over all $\binom{100}{2} = 4{,}950$ pairs and $\mathrm{SND}^{\mathrm{u}}_G(G_{0.1})$ on one Bernoulli-0.1 edge sample, with CUDA-synchronized timing. Figure 1 shows the two diversity curves overlapping throughout training while the sampled call stays near the predicted $1/p = 10\times$ speedup ($[9.0, 10.9]\times$ over all logged measurements). A separate frozen-init GPU timing sweep through $n = 500$ follows the same $\binom{n}{2}/|E|$ law: at $n = 500$, Bernoulli-0.1 reduces mean metric time from 15,502.8 ms to 1,553.6 ms (Figure 2). Together, the online $n = 100$ run and the frozen $n = 500$ sweep separate the two questions a reviewer should care about: the sampled metric tracks full SND during learning, and the wall-clock reduction persists at the larger team sizes where the quadratic complete graph becomes expensive.

6.2 Closed-loop diversity control on VMAS Dispersion

We integrate Graph-SND into the DiCo diversity controller [Bettini et al., 2024a] on VMAS Dispersion. An $n = 10$ study shows that k-NN, Bernoulli-0.1, and random d-regular Graph-SND all track $\mathrm{SND}_{\mathrm{des}} = 0.1$ and match the full-SND controller's reward within seed variation while reducing per-call metric cost (Figure 3). The stronger closed-loop evidence is the $n = 50$ head-to-head in Table 1: each of three seeds and three set points is trained twice, once with Bernoulli-0.1 Graph-SND and once with full SND. Bernoulli-0.1 tracks all targets to sub-1% relative error, yields paired reward differences centered near zero, and reduces metric cost by 9.24–9.63$\times$, matching the $10\times$ pair-count prediction. This is the main closed-loop substitution test: full SND is not merely evaluated

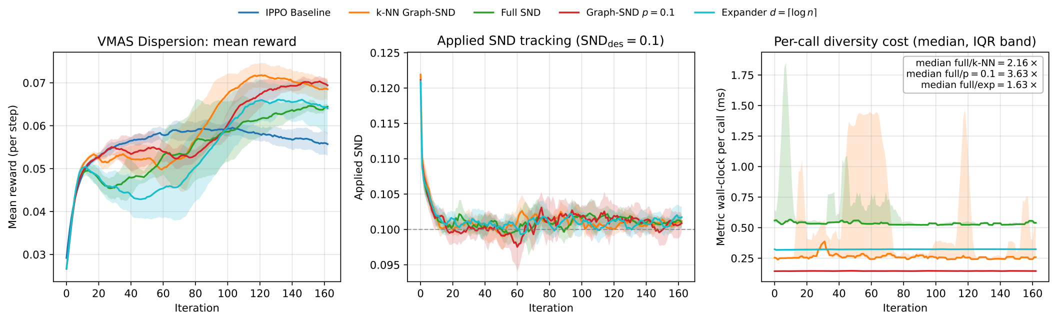

Figure 3: DiCo with Graph-SND on VMAS Dispersion (n=10, ten foods, $\mathrm{SND_{des}}$=0.1, three seeds). Sparse k-NN, Bernoulli-0.1, and expander aggregators track the full-SND controller while reducing per-call metric cost.

$\mathrm{SND_{des}}$	Rel. tracking err. (Bern/full)	Reward (Bern/full)	Metric speedup
0.12	0.53% / 0.37%	0.295±0.016 / 0.325±0.008	9.63×
0.14	0.48% / 0.31%	0.338±0.008 / 0.327±0.007	9.39×
0.15	0.42% / 0.35%	0.350±0.005 / 0.333±0.009	9.24×

Table 1: Compact main-text summary of the n=50 DiCo head-to-head against full SND. Each row aggregates three matched seeds on the late window (last 50 iterations); Bernoulli-0.1 uses $\mathbb{E}[|E|] = 122.5$ sampled edges versus 1,225 full pairs. Full curves and per-cell statistics appear in Appendix D; across all nine matched cells, the paired mean reward difference (Bernoulli minus full) is $-6.2\times10^{-4} \pm 9.2\times10^{-3}$ SEM. The largest single-cell mean gap is $\sim$9% at $\mathrm{SND_{des}}$=0.12, offset by Bernoulli-0.1 leading at the two higher set points; the paired analysis averages over this variation.

post hoc, but trained as the competing DiCo signal on the same seed and set-point grid. A separate post-hoc full-SND audit of the trained actions (Table 2 and Appendix D, Figure 10) confirms that the Bernoulli-controlled policies' complete-graph SND also stays within 0.61% of the target; this audit signal is never fed back into the controller. The only systematic difference is the aggregation used by the controller, and the observed differences are smaller than seed-to-seed variation.

6.3 Expander sparsification ablation

We test the deterministic fixed-G theory by comparing random d-regular graphs to Bernoulli, uniform-size, and k-NN graphs at matched edge counts $d \in \{\lceil \log_2 n \rceil, \lceil 2\log_2 n \rceil, \lceil 4\log_2 n \rceil\}$, $n \in \{50, 100, 200, 500\}$, five graph seeds each. Figure 4 shows the load-bearing result: d-regular expanders keep $\mathrm{SND}_G^{\mathrm{u}}/\mathrm{SND}$ in $[0.9987, 1.0013]$ at every scale, including $n = 500$ with $d = 9$ and only 2,250 of 124,750 pairs. Their edge forwarding index follows the predicted $\mathcal{O}(n \log n/d)$ scaling and is 2–39× lower than matched-edge baselines at $n = 500$, explaining why the forwarding-index bound favors expanders even though the worst-case $\mathcal{O}(\log n)$ ratio is conservative. The trained $n = 100$ checkpoint panel preserves the same ordering (Appendix E). A non-Gaussian PettingZoo MPE simple-spread transfer panel with categorical policies and TVD distance also shows small bias and matched cost reduction (Appendix C, Table 5).

The expander claim is deliberately split into worst-case and empirical parts. The forwarding-index theorem gives a deterministic guarantee for arbitrary metric distance tables, not a promise of near-unit distortion. The near-unit ratios in Figure 4 are a data-dependent observation, partially explained by Proposition 17: when the realized distance matrix has controlled normalized nuclear norm, expansion turns the logarithmic worst-case dependence into a decreasing spectral discrepancy bound. This distinction matters because fixed sparse graphs change the measurement target unless their structure is justified; the expander result shows one fixed-graph choice that remains accurate on the tested MARL distance matrices while retaining near-linear edge count.

$\mathrm{SND}_{\mathrm{des}}$	Post-hoc full-SND err. (Bern/full)	Post-hoc minus applied (Bern/full)
0.12	0.50% / 0.48%	$2.9{\times}10^{-4}$ / $3.0{\times}10^{-4}$
0.14	0.61% / 0.36%	$4.7{\times}10^{-4}$ / $2.6{\times}10^{-4}$
0.15	0.43% / 0.46%	$3.1{\times}10^{-4}$ / $4.7{\times}10^{-4}$

Table 2: Post-hoc complete-graph SND audit for the $n{=}50$ DiCo head-to-head. Each cell is remeasured with full SND on the scaled actions at every iteration, but the audit value is not used by the controller. Bernoulli-0.1 remains within 0.61% of the desired complete-graph diversity at all set points, and the audit differs from the online applied signal by less than $5{\times}10^{-4}$.

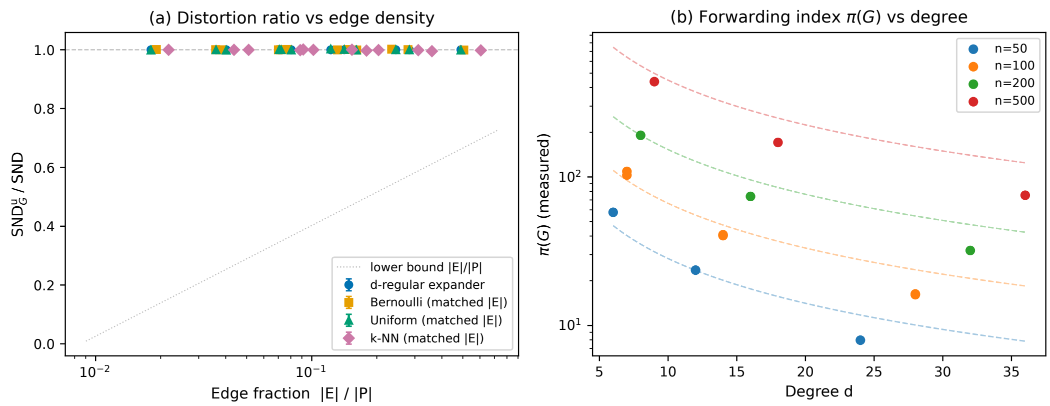

Figure 4: Expander sparsification ablation. Random d-regular graphs give the tightest near-unit $\mathrm{SND}^{\mathrm{u}}_{G}/\mathrm{SND}$ ratios at matched edge budgets, and their measured forwarding index tracks the $\mathcal{O}(n \log n/d)$ mechanism in Theorem 12.

7 Discussion

Graph-SND is best understood as a choice of graph semantics. Use Bernoulli-p when full SND is the target and sparsity is purely computational; use k-NN or communication graphs when local interactions define the intended diversity signal; use random d-regular expanders when a fixed near-linear aggregator with strong worst-case structure is desired. The same graph layer applies beyond Gaussian Wasserstein SND whenever the pairwise dissimilarity satisfies the relevant boundedness or metric assumptions (Appendix C, Table 5).

Goal	Recommended G	Main guarantee and use case
Estimate full SND	Bernoulli-p or fixed-size random sample	Horvitz–Thompson unbiasedness and $O(1/\sqrt{m})$ concentration; best when full-system diversity remains the target.
Measure local diversity	Communication, spatial, or k-NN graph	Interpretable fixed-G signal over interacting pairs; appropriate when non-edges are intentionally excluded from the semantic target.
Use a fixed sparse proxy	Random d-regular expander, $d = \Theta(\log n)$	Forwarding-index worst-case structure plus near-unit empirical ratios at $\Theta(n \log n)$ edges in our ablation.

Table 3: Practical graph-selection rule. The graph is not a nuisance parameter: it decides whether Graph-SND is estimating full SND or defining a different, localized diversity functional.

Why aggregation is the right intervention. Graph-SND is most useful when SND is not a one-off diagnostic but a quantity repeatedly evaluated inside training, selection, or diversity control. In that setting full SND imposes a quadratic tax at every metric call, while sparse aggregation exposes an explicit accuracy–cost knob: p or m for random estimators, k for local neighborhoods, and d for expander proxies. Because the intervention is only the aggregation layer, the downstream optimizer and pairwise distance remain unchanged. The evidence is therefore deliberately staged:

exact recovery and unbiasedness establish equivalence on the complete target, scaling experiments show the expected cost law, and DiCo verifies that the same replacement remains stable when the metric is used as feedback rather than only measured offline.

Failure modes and interpretation. The main failure mode is transparent from the graph semantics. A sparse fixed graph can hide diversity on non-edges: for example, if agents form two behaviorally distinct clusters but G contains mostly within-cluster edges, SND_G may be small while complete-graph SND is large. This is not a failure of the estimator theory; it means the graph has selected a local measurement target. Conversely, Bernoulli and fixed-size random graphs treat omitted edges as missing samples from the complete pair set, so their guarantees speak directly about full SND. This separation is why we evaluate fixed expanders as deterministic proxies, evaluate random graphs by unbiasedness and concentration, and audit the DiCo controller post hoc with full complete-graph SND. The same implementation can therefore support two scientifically different uses: measuring diversity only where interaction structure says it matters, or approximating the global SND metric when all-pairs aggregation is the bottleneck.

Limitations and future work. The empirical scope remains intentionally focused. Online training is shown at $n = 100$ for PPO and at $n = 50$ for DiCo, while $n = 500$ is a frozen-policy timing and expander-measurement scale. All RL runs use Independent PPO, with centralized-critic algorithms left for future work. Non-Gaussian validation is limited to one PettingZoo MPE TVD/categorical panel and synthetic checks, even though the main estimator guarantees are distance-agnostic under boundedness. Finally, Proposition 17 explains part of the gap between worst-case expander distortion and observed near-unit ratios via nuclear-norm control, and Proposition 19 gives an unconditional $\widetilde{\mathcal{O}}(1/\sqrt{n})$ probabilistic bound for the random d-regular sampling distribution; the residual gap to the empirically observed $\mathcal{O}(1/n)$-scale ratios is consistent with standard sub-Gaussian-vs-Bernstein looseness and is not closed by these tools. Natural next steps are learned or task-adaptive graphs, larger discrete-action benchmarks, and broader DiCo studies over tasks, set points, and sampling schedules.

8 Conclusion

Graph-SND turns SND's all-pairs average into a graph-edge aggregate. This single substitution unifies local diversity measurement, unbiased sparse estimation, and expander-based deterministic approximation, with theory and experiments showing the expected cost reduction while preserving the behavior of SND in training and DiCo control. The practical knob is the graph G: it decides whether the diversity signal is local by design or a cheaper approximation of the global complete-graph metric. The implication is that all-pairs aggregation should be a deliberate choice, not the default paid at every scale: when global SND is required, random sampling gives estimator guarantees; when task structure matters, fixed graphs give a sharper local signal; and when deterministic sparse proxies are desired, expanders provide a strong near-linear candidate. Code will be released upon acceptance (anonymous supplementary material accompanies this submission).

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A Proofs

We use the conventions of Section 3: $\mathcal{A}$ is the agent set with $|\mathcal{A}| = n$; $\mathcal{P} = \binom{\mathcal{A}}{2}$ is the set of unordered agent pairs, with $|\mathcal{P}| = \binom{n}{2}$; $d(i,j) \geq 0$ is the Monte Carlo behavioral distance (1), treated as a deterministic scalar given the rollout set $\mathcal{B}$; and $G = (\mathcal{A}, E, w)$ has non-negative edge weights w_{ij} with total weight $W(G) = \sum_{\{i,j\}\in E} w_{ij} > 0$.

Proof of Proposition 2 (Recovery)

If $G = K_n$ with $w_{ij} = 1$ for all $\{i,j\} \in \mathcal{P}$, then $E = \mathcal{P}$ and $W(G) = |\mathcal{P}| = \binom{n}{2}$. Substituting into (3),

$$\mathrm{SND}_G(\mathbf{D}, K_n) \;=\; \frac{1}{\binom{n}{2}} \sum_{\{i,j\}\in\mathcal{P}} 1 \cdot d(i,j) \;=\; \binom{n}{2}^{-1} \sum_{i<j} d(i,j) \;=\; \mathrm{SND}(\mathbf{D}). \quad \square$$

Proof of Proposition 3 (Non-negativity)

Each term $w_{ij}d(i,j)$ in (3) is a product of non-negative factors, so the numerator $\sum_E w_{ij}d(i,j) \geq 0$. Since $W(G) > 0$ by assumption, $\mathrm{SND}_G(\mathbf{D}, G) \geq 0$.

For the equality condition, $\mathrm{SND}_G = 0$ iff $\sum_{\{i,j\}\in E} w_{ij}d(i,j) = 0$. A sum of non-negative terms is zero iff each term is zero, so the equality holds iff $w_{ij}d(i,j) = 0$ for every $\{i,j\} \in E$. Equivalently, $d(i,j) = 0$ for every edge $\{i,j\} \in E$ with $w_{ij} > 0$. $\square$

Proof of Proposition 5 (Automorphism invariance)

Let $\sigma : \mathcal{A} \to \mathcal{A}$ be an automorphism of (G, w), meaning σ is a bijection satisfying $\{i,j\} \in E \iff \{\sigma(i), \sigma(j)\} \in E$ and $w_{ij} = w_{\sigma(i)\sigma(j)}$ for all $\{i,j\} \in E$. Then

$$\mathrm{SND}_G(\sigma \cdot \mathbf{D}, G) \;=\; \frac{1}{W(G)} \sum_{\{i,j\}\in E} w_{ij}\, d(\sigma(i), \sigma(j)).$$

Reindex the sum by $\{i', j'\} = \{\sigma(i), \sigma(j)\}$. Because σ is an automorphism, the map $\{i,j\} \mapsto \{\sigma(i), \sigma(j)\}$ is a bijection $E \to E$, and $w_{ij} = w_{\sigma(i)\sigma(j)} = w_{i'j'}$. Hence

$$\mathrm{SND}_G(\sigma \cdot \mathbf{D}, G) \;=\; \frac{1}{W(G)} \sum_{\{i',j'\}\in E} w_{i'j'}\, d(i', j') \;=\; \mathrm{SND}_G(\mathbf{D}, G). \quad \square$$

Proof of Proposition 6 (Complexity)

Evaluating (3) on a precomputed graph G reduces to computing $d(i,j)$ for each $\{i,j\} \in E$ and forming a weighted sum. The sum costs $O(|E|)$ floating-point operations, dominated by the $|E|$ pairwise-distance computations. SND in (2) instead requires $|\mathcal{P}| = \binom{n}{2}$ such computations. Since each pairwise-distance computation has the same cost in both cases, the speedup factor in the number of pairwise-distance evaluations is $\binom{n}{2}/|E|$. For k-NN graphs $|E| = O(nk)$, giving a factor of $\Theta(n/k)$. This statement does not include the cost of constructing G. $\square$

Proof of Proposition 7 (Unbiasedness)

For each $\{i,j\} \in \mathcal{P}$, let $Z_{ij} = \mathbf{1}[\{i,j\} \in E(G_p)]$. By construction, $\{Z_{ij}\}_{\{i,j\}\in\mathcal{P}}$ are independent Bernoulli(p) random variables. With $w_{ij} = 1/p$ on edges of G_p,

$$\widehat{\mathrm{SND}}_{\mathrm{HT}}(G_p) \;=\; |\mathcal{P}|^{-1} \sum_{\{i,j\}\in E(G_p)} w_{ij}\, d(i,j) \;=\; |\mathcal{P}|^{-1} \sum_{\{i,j\}\in\mathcal{P}} Z_{ij} \cdot \tfrac{1}{p} \cdot d(i,j).$$

Taking expectation and using $\mathbb{E}[Z_{ij}] = p$,

$$\mathbb{E}_{G_p}\Big[\widehat{\mathrm{SND}}_{\mathrm{HT}}(G_p)\Big] \;=\; |\mathcal{P}|^{-1} \sum_{\{i,j\}\in\mathcal{P}} p \cdot \tfrac{1}{p} \cdot d(i,j) \;=\; |\mathcal{P}|^{-1} \sum_{\{i,j\}\in\mathcal{P}} d(i,j) \;=\; \mathrm{SND}(\mathbf{D}). \quad \square$$

Proof of Theorem 9 (Concentration)

We first establish that, conditional on $|E(G_p)| = m$, $E(G_p)$ is a uniform random subset of $\mathcal{P}$ of size m.

Lemma 16. *Let $\{Z_\alpha\}_{\alpha\in\mathcal{P}}$ be i.i.d. Bernoulli(p) with $p \in (0,1)$, and let $E = \{\alpha : Z_\alpha = 1\}$. Conditional on $|E| = m$ with $0 \leq m \leq |\mathcal{P}|$, the set E is uniformly distributed over subsets of $\mathcal{P}$ of size m.*

Proof. For any fixed $T \subseteq \mathcal{P}$ with $|T| = m$,

$$\mathbb{P}(E = T \mid |E| = m) \;=\; \frac{\mathbb{P}(E = T)}{\mathbb{P}(|E| = m)} \;=\; \frac{p^m(1-p)^{|\mathcal{P}|-m}}{\binom{|\mathcal{P}|}{m} p^m (1-p)^{|\mathcal{P}|-m}} \;=\; \binom{|\mathcal{P}|}{m}^{-1},$$

which does not depend on T. $\square$

Now fix $m \geq 1$ and condition on $|E(G_p)| = m$. By Lemma 16, $E(G_p)$ is a uniform random sample of size m without replacement from the finite population

$$\Pi \ := \ \big\{d(i,j) : \{i,j\} \in \mathcal{P}\big\},$$

where $|\Pi| = |\mathcal{P}|$ and each $d(i,j) \in [0, D_{\max}]$. The population mean is

$$\mu \ := \ |\mathcal{P}|^{-1} \sum_{\{i,j\} \in \mathcal{P}} d(i,j) \ = \ \mathrm{SND}(\mathbf{D}),$$

and the sample mean is precisely

$$\mathrm{SND}^{\mathrm{u}}_G(\mathbf{D}, G_p) \ = \ m^{-1} \sum_{\{i,j\} \in E(G_p)} d(i,j).$$

Hoeffding [1963, Section 6, Theorem 4] establishes that the two-sided Hoeffding inequality applies to sampling without replacement from a bounded finite population: for any $t > 0$,

$$\mathbb{P}\big(|\mathrm{SND}^{\mathrm{u}}_G - \mathrm{SND}| \geq t \ \big| \ |E(G_p)| = m\big) \ \leq \ 2\exp\!\left(-\frac{2mt^2}{D^2_{\max}}\right). \tag{6}$$

Setting the right-hand side equal to δ and solving for t gives

$$t \ = \ D_{\max}\sqrt{\frac{\log(2/\delta)}{2m}}.$$

Thus with probability at least $1 - \delta$ conditional on $|E(G_p)| = m$,

$$|\mathrm{SND}^{\mathrm{u}}_G(\mathbf{D}, G_p) - \mathrm{SND}(\mathbf{D})| \ \leq \ D_{\max}\sqrt{\frac{\log(2/\delta)}{2m}}. \quad \square$$

Deterministic fixed-G distortion

For a connected unit-weight graph $G = (\mathcal{A}, E)$, a *path system* is an assignment of a simple i-j path $\mathcal{R}_{ij} \subseteq E$ to every pair $\{i,j\} \in \mathcal{P}$, with $\mathcal{R}_{ij} = \{\{i,j\}\}$ whenever $\{i,j\} \in E$. The *congestion* of a path system $\mathcal{R}$ at an edge $\{a,b\} \in E$ is $c_{\mathcal{R}}(a,b) := |\{\{i,j\} \in \mathcal{P} : \{a,b\} \in \mathcal{R}_{ij}\}|$, and the *edge forwarding index* of G is $\pi(G) := \min_{\mathcal{R}} \max_{\{a,b\} \in E} c_{\mathcal{R}}(a,b)$ [Chung et al., 1987, Heydemann et al., 1989]. The index depends on G alone, satisfies $\pi(K_n) = 1$, admits closed forms for many graph families (hypercubes, tori, Cayley graphs), and is $\mathcal{O}(n \log n / d)$ for d-regular graphs with bounded spectral gap [Leighton and Rao, 1999].

Proposition 17 (Spectral discrepancy under nuclear-norm control)**.** *Let G be a connected simple d-regular graph on n vertices with adjacency matrix A and $\lambda_2 := \max\{|\lambda_i(A)| : i \geq 2\}$, where $d = \lambda_1(A)$ is the trivial eigenvalue. Let $\mathbf{D} \in \mathbb{R}^{n \times n}$ be symmetric with zero diagonal. Then*

$$|\mathrm{SND}^{\mathrm{u}}_G(\mathbf{D}, G) - \mathrm{SND}(\mathbf{D})| \leq \frac{\lambda_2 + d/(n-1)}{n\,d}\,\|\mathbf{D}\|_*, \tag{7}$$

where $\|\cdot\|_*$ denotes the nuclear norm. If $\mathrm{SND}(\mathbf{D}) > 0$ and

$$\rho_*(\mathbf{D}) := \frac{\|\mathbf{D}\|_*}{n\,\mathrm{SND}(\mathbf{D})},$$

then

$$\left|\frac{\mathrm{SND}^{\mathrm{u}}_G(\mathbf{D}, G)}{\mathrm{SND}(\mathbf{D})} - 1\right| \leq \frac{\lambda_2 + d/(n-1)}{d}\,\rho_*(\mathbf{D}). \tag{8}$$

In particular, if G is Ramanujan ($\lambda_2 \leq 2\sqrt{d-1}$) and $\rho_*(\mathbf{D}) \leq C$, then the relative error is at most $C(2\sqrt{d-1} + d/(n-1))/d = \mathcal{O}(C/\sqrt{d})$; at $d = \Theta(\log n)$ this is $\mathcal{O}(C/\sqrt{\log n})$.

Remark 18 (What the spectral bound explains). Section 6.3 measures $\mathrm{SND}^{\mathrm{u}}_G/\mathrm{SND} \in [0.9987, 1.0013]$ on random d-regular graphs at $d{=}\Theta(\log n)$, far tighter than the worst-case $\mathcal{O}(\log n)$ ratio of Corollary 13. Proposition 17 formalizes a distribution-dependent explanation: if the realized pairwise dissimilarity matrix has controlled normalized nuclear norm $\rho_*(\mathbf{D})$, then spectral expansion improves the deterministic bound from logarithmic distortion to a decreasing

$\mathcal{O}(\rho_*(\mathbf{D})/\sqrt{\log n})$ relative-error bound at $d = \Theta(\log n)$. This is still a fixed-G, worst-case alignment bound over matrices satisfying the nuclear-norm condition; it does not fully explain the much smaller empirical ratios, which likely also use random-graph variance reduction and the specific distributional structure of MARL distance matrices. Proposition 19 below gives an unconditional probabilistic sharpening for random d-regular graphs; closing the remaining gap to the empirically observed $\mathcal{O}(1/n)$-scale ratios would require sharper distribution-dependent or Bernstein-type analysis.

Proof of Proposition 17. Let $J = \mathbf{1}\mathbf{1}^\top$ and define the discrepancy matrix

$$M_G := A - \frac{d}{n-1}(J - I).$$

For the all-ones vector, $A\mathbf{1} = d\mathbf{1}$ and $(J - I)\mathbf{1} = (n-1)\mathbf{1}$, so $M_G\mathbf{1} = 0$. For any eigenvector $v \perp \mathbf{1}$ of A with $Av = \lambda v$, we have $Jv = 0$ and $(J - I)v = -v$, hence

$$M_G v = \left(\lambda + \frac{d}{n-1}\right) v.$$

Therefore

$$\|M_G\|_{\mathrm{op}} \le \lambda_2 + \frac{d}{n-1}.$$

Using $D_{ii} = 0$, $|E| = nd/2$, and $|\mathcal{P}| = n(n-1)/2$,

$$\langle M_G, \mathbf{D}\rangle_F = 2|E|\,\mathrm{SND}^{\mathrm{u}}_G(\mathbf{D}, G) - \frac{d}{n-1}\,2|\mathcal{P}|\,\mathrm{SND}(\mathbf{D}) = nd\big(\mathrm{SND}^{\mathrm{u}}_G(\mathbf{D}, G) - \mathrm{SND}(\mathbf{D})\big).$$

Operator-norm/nuclear-norm duality gives $|\langle M_G, \mathbf{D}\rangle_F| \le \|M_G\|_{\mathrm{op}}\|\mathbf{D}\|_*$, proving (7). Dividing by $\mathrm{SND}(\mathbf{D})$ proves (8); the Ramanujan specialization substitutes $\lambda_2 \le 2\sqrt{d-1}$.

Proof of Theorem 12. *Lower bound.* Since $d(i, j) \ge 0$, dropping the $|\mathcal{P}| - |E|$ non-edge terms from the full sum cannot increase it:

$$|\mathcal{P}| \cdot \mathrm{SND}(\mathbf{D}) \;=\; \sum_{\{i,j\}\in\mathcal{P}} d(i, j) \;\ge\; \sum_{\{i,j\}\in E} d(i, j) \;=\; |E| \cdot \mathrm{SND}^{\mathrm{u}}_G(\mathbf{D}, G),$$

with equality iff $d(i, j) = 0$ for every non-edge $\{i, j\} \in \mathcal{P} \setminus E$.

Upper bound. Fix a path system $\mathcal{R}$ on G in the sense of Section 5. For each edge pair $\{i, j\} \in E$, the single-edge path $\mathcal{R}_{ij} = \{\{i, j\}\}$ gives the identity $d(i, j) = \sum_{\{a,b\}\in\mathcal{R}_{ij}} d(a, b)$. For each non-edge pair $\{i, j\} \in \mathcal{P} \setminus E$, writing the path as $\mathcal{R}_{ij} = (i = v_0, v_1, \ldots, v_L = j)$ and applying the triangle inequality $L - 1$ times,

$$d(i, j) \;\le\; d(v_0, v_1) + d(v_1, v_L) \;\le\; \cdots \;\le\; \sum_{k=0}^{L-1} d(v_k, v_{k+1}) \;=\; \sum_{\{a,b\}\in\mathcal{R}_{ij}} d(a, b).$$

Combining both cases and exchanging the order of summation,

$$\sum_{\{i,j\}\in\mathcal{P}} d(i, j) \;\le\; \sum_{\{i,j\}\in\mathcal{P}} \sum_{\{a,b\}\in\mathcal{R}_{ij}} d(a, b) \;=\; \sum_{\{a,b\}\in E} c_{\mathcal{R}}(a, b)\, d(a, b),$$

where $c_{\mathcal{R}}(a, b) := |\{\{i, j\} \in \mathcal{P} : \{a, b\} \in \mathcal{R}_{ij}\}|$ is the congestion of $\mathcal{R}$ at edge $\{a, b\}$. Since $d(a, b) \ge 0$,

$$\sum_{\{a,b\}\in E} c_{\mathcal{R}}(a, b)\, d(a, b) \;\le\; \Big(\max_{\{a,b\}\in E} c_{\mathcal{R}}(a, b)\Big) \sum_{\{a,b\}\in E} d(a, b).$$

This inequality holds for every path system $\mathcal{R}$, so it holds in particular for one that attains the minimum in the definition of $\pi(G)$, for which the bracket equals $\pi(G)$. Therefore

$$|\mathcal{P}| \cdot \mathrm{SND}(\mathbf{D}) \;\le\; \pi(G) \cdot \sum_{\{a,b\}\in E} d(a, b) \;=\; \pi(G) \cdot |E| \cdot \mathrm{SND}^{\mathrm{u}}_G(\mathbf{D}, G).$$

Dividing by $|\mathcal{P}|$ yields $\mathrm{SND}(\mathbf{D}) \le (|E|\pi(G)/|\mathcal{P}|)\,\mathrm{SND}^{\mathrm{u}}_G(\mathbf{D}, G)$. The recovery statement follows because K_n satisfies $\pi(K_n) = 1$ (take $\mathcal{R}_{ij} = \{\{i, j\}\}$ for all pairs) and $|E| = |\mathcal{P}| = \binom{n}{2}$. $\square$

Fractional sharpening. Replacing the integral path system by a convex combination of paths $f_{ij} = \sum_{\mathcal{R}\in\Pi_{ij}} \alpha_{\mathcal{R}} \mathbf{1}_{\mathcal{R}}$ ($\alpha_{\mathcal{R}} \geq 0$, $\sum_{\mathcal{R}} \alpha_{\mathcal{R}} = 1$) and repeating the argument with the fractional congestion $C^*_{\mathbf{f}}(a,b) := \sum_{\{i,j\}\in\mathcal{P}} f_{ij}(a,b)$ tightens the upper bound of Eq. (5) to $|E|\pi^*(G)/|\mathcal{P}|$, where $\pi^*(G) = \min_{\mathbf{f}} \max_{\{a,b\}\in E} C^*_{\mathbf{f}}(a,b) \leq \pi(G)$ is the fractional edge forwarding index, computable by a multicommodity-flow LP [Leighton and Rao, 1999].

Probabilistic distortion for random d-regular graphs

Proposition 19 (Unconditional probabilistic distortion)**.** *Let G be drawn uniformly from simple d-regular graphs on n vertices, with $d \geq 3$, nd even, and $d^2 = o(n)$ (so the headline setting $d = \Theta(\log n)$ is included). Let $\mathbf{D} \in \mathbb{R}^{n\times n}$ be symmetric with $D_{ii} = 0$ and $D_{ij} \in [0, D_{\max}]$. Then for any $\delta \in (0,1)$, with probability at least $1-\delta$ over G,*

$$\left| \mathrm{SND}^{\mathrm{u}}_G(\mathbf{D}, G) - \mathrm{SND}(\mathbf{D}) \right| \;\leq\; D_{\max} \sqrt{\frac{8}{nd}\left(\log\frac{4}{\delta} + \frac{d^2-1}{4}\right)}. \tag{9}$$

In particular, for $d = \Theta(\log n)$, this gives $|\mathrm{SND}^{\mathrm{u}}_G - \mathrm{SND}| = \widetilde{\mathcal{O}}(D_{\max}/\sqrt{n})$ with high probability, independent of any structural assumption on $\mathbf{D}$ beyond boundedness.

Proof. The argument uses three classical ingredients.

Step 1: Configuration model. Represent the uniform random d-regular multigraph via the configuration model [Bollobás, 1980]: assign d labeled stubs to each vertex and draw a uniformly random perfect matching π of the nd stubs. Define $f(\pi) := \sum_{e\in E(G(\pi))} D_e$, so $\mathrm{SND}^{\mathrm{u}}_G(\mathbf{D}, G(\pi)) = 2f(\pi)/(nd)$. By edge symmetry, $\mathbb{E}_\pi[f(\pi)] = (nd/2)\,\mathrm{SND}(\mathbf{D})$.

Step 2: McDiarmid's permutation inequality. For matchings π, π' differing by a single transposition of two stubs in different pairs, the edge sets differ in exactly four edges (two deleted, two added), so $|f(\pi) - f(\pi')| \leq 2D_{\max}$. By Hoeffding's inequality for permutations [McDiarmid, 1989, §6.1] with $N = nd$ and bounded difference $c = 2D_{\max}$:

$$\Pr_\pi\big[|f(\pi) - \mathbb{E} f(\pi)| \geq t\big] \;\leq\; 2\exp\!\left(-\frac{t^2}{2\,nd\,D_{\max}^2}\right).$$

Setting $t = (nd/2)\epsilon$ and dividing through:

$$\Pr_\pi\big[|\mathrm{SND}^{\mathrm{u}}_G - \mathrm{SND}| \geq \epsilon\big] \;\leq\; 2\exp\!\left(-\frac{nd\,\epsilon^2}{8\,D_{\max}^2}\right).$$

Step 3: Conditioning on simplicity. The uniform simple d-regular graph is the configuration multigraph conditioned on being simple. By Bollobás [1980] and McKay [1981], in the sparse regime $d^2 = o(n)$ —which contains the headline setting $d = \Theta(\log n)$ used here— $\Pr_\pi(G(\pi) \text{ is simple}) = \exp\big(-(d^2-1)/4 + o(1)\big)$ as $n \to \infty$, so $\Pr_\pi(G(\pi) \text{ is simple}) \geq \frac{1}{2} e^{-(d^2-1)/4}$ for n sufficiently large. Therefore $\Pr_{G\sim\text{simple}}[A] \leq 2e^{(d^2-1)/4} \Pr_\pi[A]$. Combining with Step 2 and inverting for δ yields (9). ☐

B Hyperparameters

Table 4 lists the Independent PPO hyperparameters used for all experiments in Section 6. All values are held fixed across team sizes $n \in \{4, 8, 16, 100\}$ except as noted below.

The $n = 100$ scaling run of Section 6.1 additionally uses 500 iterations, `num_envs` $= 32$ (unchanged), and scenario parameters `world_spawning_x` $=$ `world_spawning_y` $= 4.0$, `agent_radius` $= 0.03$ to accommodate the larger team size.

C Empirical verification of core propositions

This appendix contains the protocols, figures, and diagnostics for the three structural verifications summarized in Section 6: exact recovery on K_n (Proposition 2), unbiasedness of the Horvitz–Thompson estimator (Proposition 7), and the Hoeffding/Serfling concentration bounds (Theorem 9).

Hyperparameter	**Value**
Optimizer	Adam
Learning rate	3×10^{-4}
Clip parameter ϵ	0.2
Entropy coefficient	0.01
Value loss coefficient	0.5
Max gradient norm	0.5
GAE λ	0.95
Discount γ	0.99
Parallel environments	32
Rollout horizon	128
PPO epochs per update	4
Mini-batch size	512
Training iterations	100
Hidden layer width	64
Hidden layers	2
Activation	Tanh

Table 4: Independent PPO hyperparameters used for all experiments. The rollout batch per iteration is Rollout horizon $\times$ Parallel environments $= 128 \times 32 = 4{,}096$ transitions per agent.

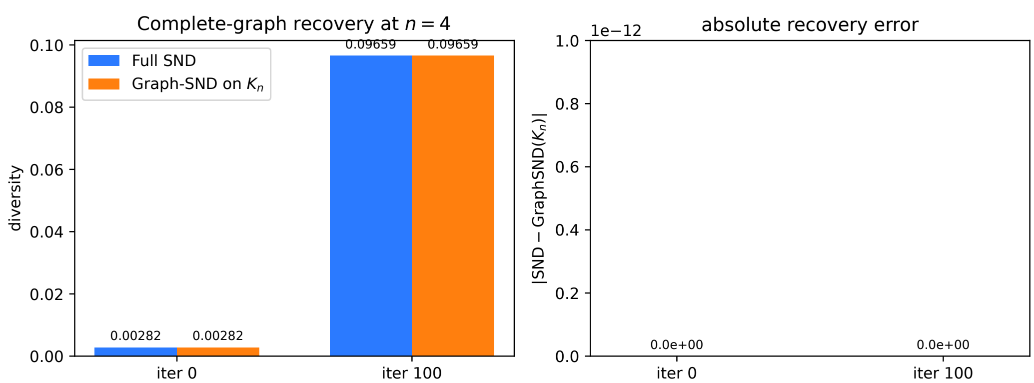

Figure 5: Proposition 2: recovery of SND by SND_G on the complete graph. Left: side-by-side values of SND and SND_G on K_4 at iter 0 (near-homogeneous policies) and iter 100 (trained policies). Right: absolute recovery error, which is 0.0 in both cases (the y-axis is shown at 10^{-12} resolution for context).

Recovery (Proposition 2). We evaluate $\mathrm{SND}(\mathbf{D})$ and $\mathrm{SND}_G(K_4)$ on the same pairwise-distance matrix at iter 0 (near-homogeneous policies) and iter 100 (trained policies). The absolute error is 0.0 in every cell; on a log axis the gap is indistinguishable from numerical noise at 10^{-12} resolution (Figure 5). This validates that our Graph-SND implementation is correctly aligned with SND as a reference for the remaining experiments.

Unbiasedness (Proposition 7). We test $\mathbb{E}_{G_p}[\widehat{\mathrm{SND}}_{\mathrm{HT}}(G_p)] = \mathrm{SND}(\mathbf{D})$ at $n{=}8$ by drawing 2,000 independent Bernoulli-p graphs for each $p \in \{0.1, 0.25, 0.5, 0.75\}$, computing $\widehat{\mathrm{SND}}_{\mathrm{HT}}$ on each, and reporting the sample mean with a 95% CI (Figure 6). The CI contains $\mathrm{SND}(\mathbf{D})$ in every cell across both checkpoints. As a sharper test, the standardized deviation $|\mathrm{bias}|/\mathrm{SE}$ is approximately $\mathcal{N}(0, 1)$ per cell under an unbiased Gaussian-error estimator; the maximum observed value across all eight (checkpoint, p) cells is 1.42, well below $|z_{0.975}| = 1.96$. Estimator variance scales as $1/p$, visible as widening CIs at small p.

Concentration (Theorem 9). We evaluate Theorem 9 at $n{=}16$ ($|\mathcal{P}|{=}120$) by drawing 2,000 uniform-size edge samples of each size $m \in \{6, 12, 24, 48, 72, 96\}$ (5%–80% of all pairs), com-

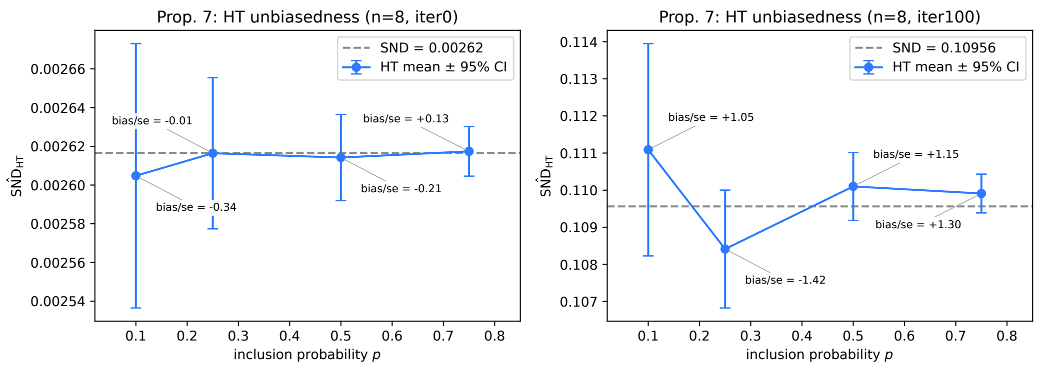

Figure 6: Proposition 7: empirical unbiasedness of $\widehat{\mathrm{SND}}_{\mathrm{HT}}(G_p)$ at $n = 8$. Blue dots and whiskers are the sample mean and 95% confidence interval over 2,000 random graphs per p; the dashed line is the true $\mathrm{SND}(\mathbf{D})$. Annotations give the standardized deviation $|\mathrm{bias}|/\mathrm{SE}$, all of which are well below 1.96. Left: iter 0 checkpoint. Right: iter 100 checkpoint.

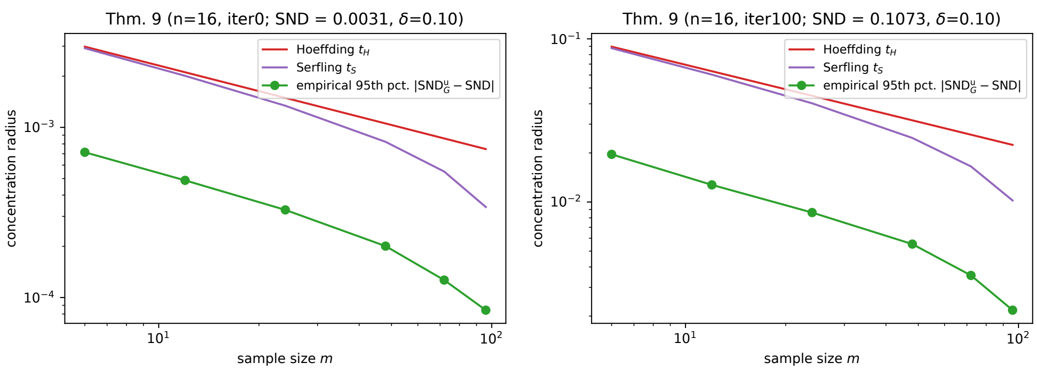

Figure 7: Theorem 9: concentration of $\mathrm{SND}^{\mathrm{u}}_G$ around SND at $n = 16$. Red: Hoeffding radius t_H from Theorem 9. Purple: Serfling radius t_S from Remark 11. Green: empirical 95th-percentile of $|\mathrm{SND}^{\mathrm{u}}_G - \mathrm{SND}|$ over 2,000 uniform-size edge samples per m. Both axes are log-scaled. Across all 12 cells, zero draws violate either bound. The empirical gap reflects the fact that Hoeffding and Serfling are worst-case over populations bounded in $[0, D_{\max}]$; the true $d(i, j)$ variance is much smaller.

puting $\mathrm{SND}^{\mathrm{u}}_G$ on each, and comparing the empirical distribution of $|\mathrm{SND}^{\mathrm{u}}_G - \mathrm{SND}|$ against theory at δ=0.10 (Figure 7). Across all 12 cells, the Hoeffding violation rate is 0.00—no draw out of 2,000 exceeds t_H—well under the allowed δ; the same holds for the Serfling refinement of Remark 11. The empirical curve sits well below both bounds, as expected for distribution-free guarantees that are worst-case tight only at population variance $D^2_{\max}/4$; it still exhibits the predicted $O(1/\sqrt{m})$ slope on the log-log axis, and Serfling is uniformly tighter than Hoeffding with the gap widening at large sampling fractions.

Non-VMAS discrete-action transfer (MPE simple-spread + TVD). To verify transfer beyond the Gaussian/VMAS regime, we run a PettingZoo MPE simple-spread panel at n=10 with discrete actions, categorical policies, and TVD distance. We train three IPPO seeds with `experiments/mpe_ippo_training.py` (500 iterations, 20 rollouts/iteration), then run `experiments/mpe_measurement_panel.py` with 2,000 draws per seed for three sparse graphs (Bernoulli-0.1, Bernoulli-0.25, expander-d2) and full SND. We report means across seeds, plus the max absolute bias over the three seeds. Table 5 shows that sparse estimators remain close to full SND (max $|\mathrm{bias}| = 2.92 \times 10^{-3}$) while reducing metric wall-clock cost by 1.97–4.87×. A random-init baseline panel (same seeds, same draw budget, no checkpoint load) shows near-zero violation

Regime	Estimator	Mean viol.	Max $\lvert\text{bias}\rvert$	Mean ms	Speedup
trained	Bernoulli-0.1	0.6252	2.917×10^{-3}	0.063	4.87×
trained	Bernoulli-0.25	0.3757	1.306×10^{-3}	0.107	2.86×
trained	Expander-$d2$	0.2302	0.703×10^{-3}	0.158	1.97×
random-init	Bernoulli-0.1	0.0043	0.326×10^{-3}	0.063	4.86×
random-init	Bernoulli-0.25	0.0000	0.184×10^{-3}	0.106	2.86×
random-init	Expander-$d2$	0.0000	0.274×10^{-3}	0.143	2.13×

Table 5: PettingZoo MPE simple-spread transfer panel (n=10, categorical policies, TVD distance, three seeds, 2,000 draws/seed). 'Mean viol.' is the fraction of draws with $|\mathrm{SND}_G^{\mathrm{u}} - \mathrm{SND}| > 0.05$ averaged across seeds—a fixed-threshold diagnostic, not the Hoeffding violation rate (which requires the population-dependent bound from Theorem 9). Because TVD lies in $[0, 1]$, the 0.05 threshold is loose relative to typical Hoeffding radii at δ=0.1; the small max $|\text{bias}|$ column is the more informative estimator-quality metric. 'Speedup' is relative to full SND wall-clock in the same regime.

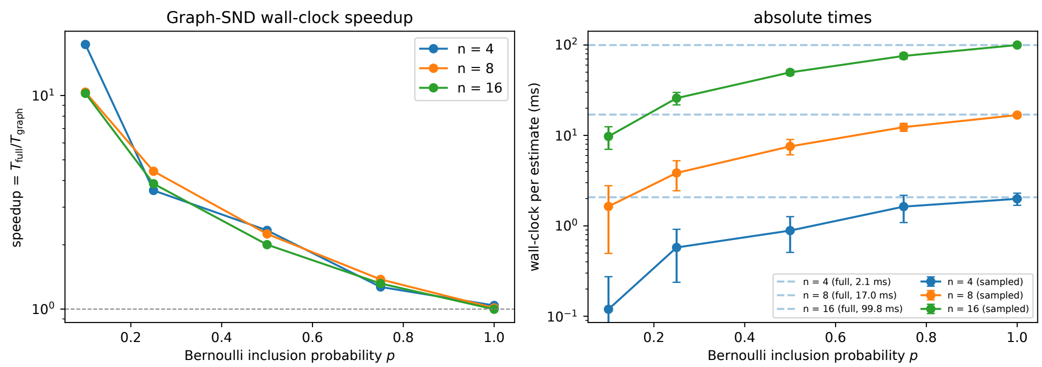

Figure 8: Proposition 6: wall-clock scaling of SND_G versus SND (CPU, $n \in \{4, 8, 16\}$). Left: speedup $T_{\text{full}}/T_{\text{sampled}}$ against Bernoulli inclusion probability p, one line per team size n. Right: absolute wall-clock time per estimate with error bars over 20 trials. Dashed horizontal lines mark the full-SND cost at each n; solid lines are the sampled SND_G cost. The red curve in the right panel of Figure 2 plots the $p = 0.1$ speedups reported here.

rates and similarly small bias. This is a transfer/robustness check for the graph aggregation layer, not a claim of task-level MPE training superiority.

CPU wall-clock sweep at $n \in \{4, 8, 16\}$. As a pre-condition to the larger-n GPU results reported in Section 6.1, we run the same SND_G-vs-SND timing comparison on the iter-100 checkpoint across $n \in \{4, 8, 16\}$ and $p \in \{0.1, 0.25, 0.5, 0.75, 1.0\}$, 20 trials per cell, CPU only. Figure 8 reports speedup and absolute times: at p=0.1 we observe 10–17× speedups across n, decaying monotonically to $\approx 1\times$ at p=1.0, consistent with the $\binom{n}{2}/\mathbb{E}[|E(G_p)|] = 1/p$ analytical prediction modulo small-n vectorization overhead.

Graph construction microbenchmark. Because Proposition 6 counts pairwise distance evaluations and excludes graph construction, we separately time the CPU-side graph builders used by the experiments. Table 6 reports median wall-clock over 50 trials for $n \in \{50, 100\}$ and 20 trials for $n = 500$ at the same edge budgets used in the expander ablation. Even exact k-NN construction remains below 5 ms at $n = 500$ in this implementation, far below the full pairwise metric time reported in Section 6.1; Bernoulli timings in Figure 2 already include sampling and edge-transfer overhead.

Graph builder	$n = 50$	$n = 100$	$n = 500$
Bernoulli	0.016 ms	0.036 ms	0.875 ms
Uniform-size sample	0.014 ms	0.022 ms	0.440 ms
Random d-regular	0.159 ms	0.345 ms	2.352 ms
Exact k-NN	0.370 ms	0.734 ms	4.091 ms

Table 6: Median CPU graph-construction time at matched edge budgets $|E| \approx nd/2$, with $d \in \{6, 7, 9\}$ for $n \in \{50, 100, 500\}$. These timings isolate graph construction only; metric-call timings are reported separately in Figure 2.

D DiCo head-to-head at n=50: Bernoulli-0.1 Graph-SND vs full SND

We extend the closed-loop experiment of Section 6.2 along three axes: team size (from n=10 to n=50, $\approx$5$\times$ more agents), the diversity set point (from the single $\mathrm{SND_{des}}$=0.1 used in the main text to three values $\mathrm{SND_{des}} \in \{0.12, 0.14, 0.15\}$), and *the choice of estimator itself*: at every $(\mathrm{seed}, \mathrm{SND_{des}})$ cell we run the same 167-iteration IPPO training twice, once with Bernoulli-0.1 Graph-SND and once with the full SND estimator (i.e. the unmodified DiCo codepath on K_n), yielding an 18-run (2$\times$3$\times$3) head-to-head at matched seeds and matched hyperparameters. Hyperparameters match Section 6.2 apart from `on_policy_n_envs_per_worker`=120 and `on_policy_collected_frames_per_batch`=12,000 (reduced from 600 and 60,000 to keep the per-iteration rollout within a single RTX 4090's memory budget at 5$\times$ more per-agent tensors); the two estimators see bit-identical rollouts and differ only in the diversity aggregation on each PPO forward. The expected edge count fed to the controller under the Bernoulli variant is $\mathbb{E}[|E(G_{0.1})|]$=$0.1\binom{50}{2}$=122.5 against $\binom{50}{2}$=1,225 under full SND (10% edge density), a nominal 10$\times$ reduction in per-call metric work.

Figure 9 summarizes the nine paired runs along four axes. Panel (a) overlays the per-iteration applied SND=$\mathrm{SND}_t\, s_t$ trajectories for the two estimators at each set point; after a $\sim$25-iteration transient every cell settles onto its target line and both estimators produce visually overlapping bands for the remaining $\sim$140 iterations of training. Panel (b) shows the corresponding per-iteration episode reward: the two estimators produce reward curves within observed seed-to-seed variation across all three set points, with higher set points yielding monotonically higher terminal reward (consistent with Dispersion rewarding role specialization among a large team). Panel (c) quantifies steady-state tracking: the late-window (50-iteration tail) relative error $|\text{applied SND} - \mathrm{SND_{des}}|/\mathrm{SND_{des}}$ is 0.42–0.53% for Bernoulli-0.1 and 0.31–0.37% for full SND, with full SND slightly tighter than Bernoulli-0.1 in every cell—expected, as full SND is the ground truth and Bernoulli-0.1 queries only $\sim$10% of pairs. Both estimators remain well under the 1% floor of the controller. Panel (d) isolates per-call metric cost: Bernoulli-0.1 averages $\sim$2.0 ms/call, full SND averages $\sim$18.9 ms/call, a 9.24–9.63$\times$ ratio that tracks the analytical $\binom{n}{2}/\mathbb{E}[|E|]$=10 prediction of Proposition 6 nearly exactly. Because full SND iters take 52.4 s end-to-end at this n and batch size—dominated by rollout collection and PPO updates rather than by the metric—the absolute wall-clock savings at n=50 are a few percent of training time; the same $\sim$10$\times$ metric-call speedup translates into larger end-to-end savings as the metric's share of per-iteration time grows with n (see Figure 2).

Table 7 reports the numerical summary aggregated across three seeds per cell. Applied SND is within 0.6% of the target for both estimators at all three set points; cross-seed standard error on applied SND is at most 1.2×10^{-4} (Bernoulli-0.1) and 1.0×10^{-4} (full). Across the nine matched $(\mathrm{seed}, \mathrm{SND_{des}})$ cells, the paired mean reward difference (Bernoulli minus full) is $-6.2{\times}10^{-4} \pm 9.2{\times}10^{-3}$ SEM, with a two-sided sign test p=1.0. The head-to-head thus upgrades the closed-loop conclusion of Section 6.2 from "matches full SND at n=10" to "matches full SND at n=50 over three distinct set points, with full SND actually trained to convergence on each cell", and does so at the 9$-$10$\times$ lower per-call metric cost predicted by the cost invariant. In other words, at n=50 the central substitution claim is demonstrated directly rather than extrapolated: DiCo with Bernoulli-0.1 Graph-SND is a drop-in replacement for DiCo with full SND.

To rule out the possibility that the sparse controller merely tracks its own sparse feedback signal, we repeat the same 18-cell grid with an audit callback that computes full complete-graph SND on the scaled actions at every iteration. This post-hoc signal is never fed back into the controller. For the Bernoulli-0.1 arm, post-hoc full-SND relative tracking error is 0.43–0.61% across the three set

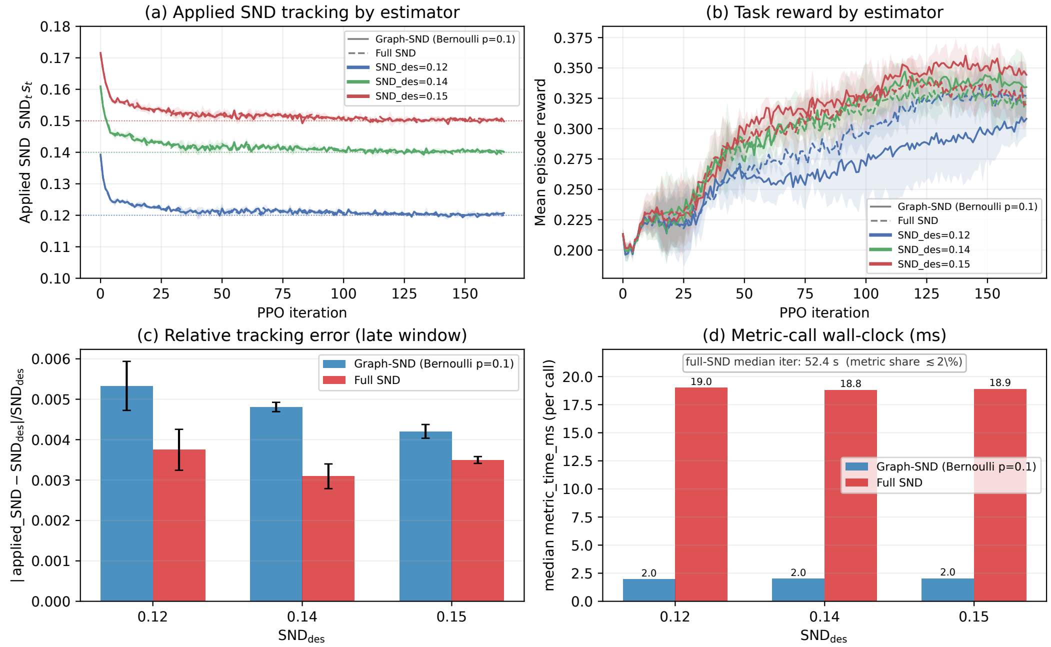

Figure 9: DiCo head-to-head at $n=50$, 3×3 grid of seeds and set points, each cell trained twice with matched rollouts (one Bernoulli-0.1 Graph-SND, one full SND), 18 runs total. **(a)** Applied $\mathrm{SND}=\mathrm{SND}_t\ s_t$ per iteration; color encodes $\mathrm{SND}_{\mathrm{des}} \in \{0.12, 0.14, 0.15\}$, solid lines are Bernoulli-0.1, dashed lines are full SND; bands are ±1 std across seeds. **(b)** Mean episode reward on the same runs, same color/line encoding. **(c)** Late-window (50-iter tail) relative tracking error; full SND is slightly tighter than Bernoulli-0.1 but both remain well under 1%. **(d)** Median per-call metric wall-clock; full SND median iter-time of $52.4\,\mathrm{s}$ is annotated to contextualise the $\sim9.5\times$ metric speed-up (which is $\lesssim2\%$ of per-iter time at $n=50$ but grows with n).

Table 7: DiCo at $n=50$ head-to-head, aggregated across three seeds on the late window (last 50 iterations). "Applied SND" and "reward" are mean $\pm$ cross-seed standard error (SEM); "rel. track. err." is mean absolute tracking error divided by $\mathrm{SND}_{\mathrm{des}}$; "metric time" is the cross-cell median of per-run median per-call times for the named estimator. Bernoulli-0.1 has expected edge count 122.5 out of $\binom{50}{2}=1{,}225$; full SND evaluates all $1{,}225$ pairs.

	Applied SND		Reward		Rel. tracking err.		Metric time (ms)	
$\mathrm{SND}_{\mathrm{des}}$	Bern-0.1	Full	Bern-0.1	Full	Bern-0.1 (%)	Full (%)	Bern-0.1	Full
0.12	0.12032±0.00012	0.12026±0.00010	0.295±0.016	0.3251±0.0080	0.53	0.37	1.97	19.0
0.14	0.140354±0.000073	0.140166±0.000025	0.3384±0.0083	0.3268±0.0072	0.48	0.31	2.00	18.8
0.15	0.150261±0.000011	0.150301±0.000015	0.3498±0.0052	0.3329±0.0086	0.42	0.35	2.04	18.9

points; for the full-SND arm it is 0.36–0.48%. The mean post-hoc complete-graph SND exceeds the online applied signal by only 2.9×10^{-4}–4.7×10^{-4}, confirming that the sparse-controlled policies also match the intended full-SND diversity level.

E Expander ablation: supplementary panels

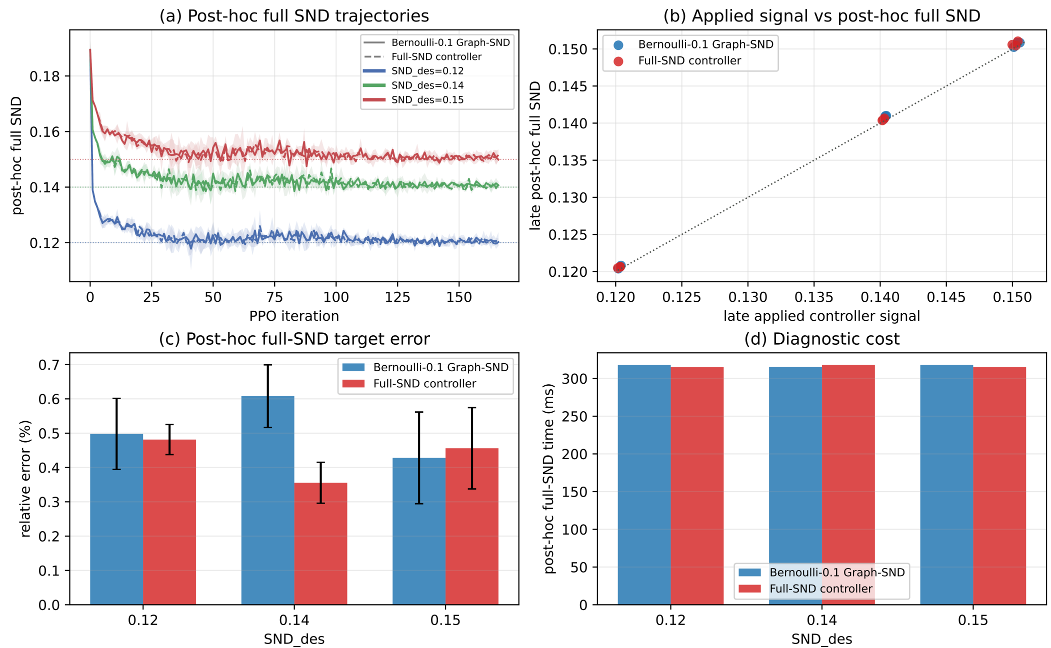

Figure 10: Post-hoc complete-graph SND audit for the $n{=}50$ DiCo head-to-head. The audit computes full SND on the scaled actions every iteration while leaving the online controller unchanged. Bernoulli-0.1 tracks the desired diversity under this complete-graph measurement to within 0.61% relative error, so the sparse controller is not merely matching its own sparse feedback signal.

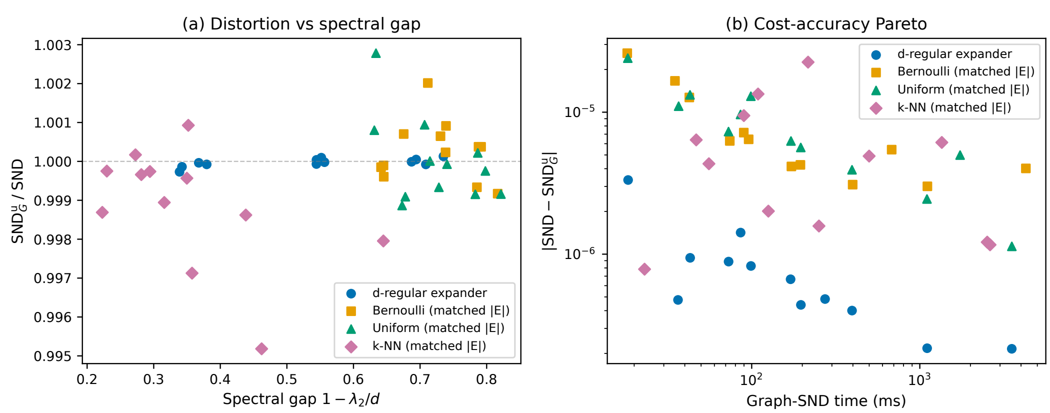

Figure 11: Continuation of Figure 4 with additional panels. **Expander sparsification ablation (supplementary).** **(a)** Empirical ratio $\mathrm{SND}^{\mathrm{u}}_G/\mathrm{SND}$ vs spectral gap $1{-}\lambda_2/d$ across four graph families. **(b)** Absolute distortion $|\mathrm{SND}{-}\mathrm{SND}^{\mathrm{u}}_G|$ vs graph-SND wall-clock time, showing the cost–accuracy Pareto frontier.

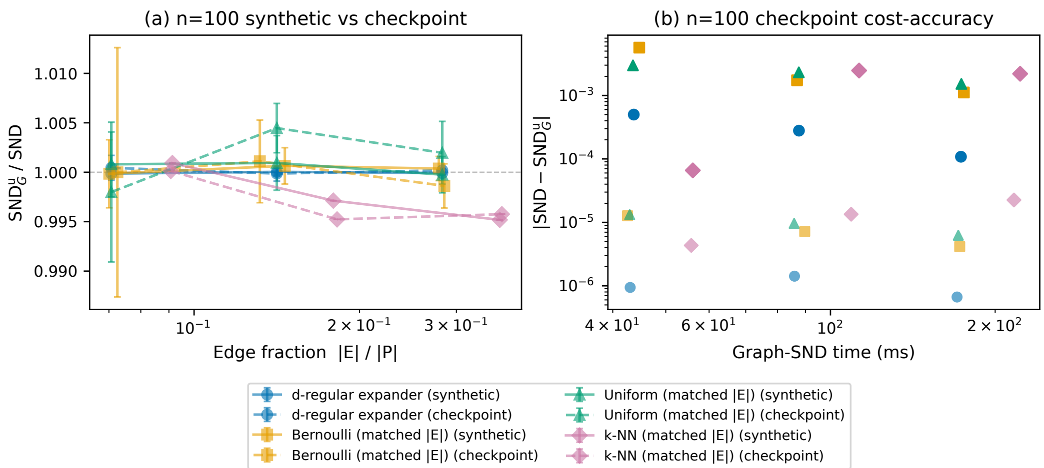

Figure 12: Continuation of Figure 4 with additional panels. **Expander ablation, trained-policy transfer at n=100 (supplementary).** **(a)** At matched edge budgets, the distortion ratio $\mathrm{SND}^{\mathrm{u}}_{G}/\mathrm{SND}$ for synthetic rollouts (solid) and a trained checkpoint rollout (dashed) remains near 1, with the d-regular family staying most tightly concentrated around unit ratio. **(b)** Cost–accuracy points for synthetic vs trained rollouts: absolute distortion shifts upward under the larger trained-policy SND scale, while the graph-family ordering and timing structure remain consistent with the structural mechanism of Theorem 12.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and Section 1 state the method, theory, scaling, expander, and DiCo claims made by the paper. The scope restrictions (IPPO only, VMAS Multi-Agent Goal Navigation and Dispersion plus one MPE simple-spread transfer panel at $n{=}10$ with categorical policies/TVD; online training at $n{=}100$ for PPO and $n{=}50$ for DiCo; $n{=}500$ used for frozen-policy timing and expander measurement) are stated in Section 7 (Limitations) and do not contradict any claim in the abstract or introduction.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 7 contains a dedicated "Limitations" paragraph itemizing five scope limits (scale/timing regime coverage, scenario and DiCo-configuration coverage, fixed-G ablation, narrow non-Gaussian coverage, IPPO-only scope).

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [Yes]

   Justification: The main-text formal results (Propositions 2, 3, 5, 6, 7, Theorem 9, Theorem 12, and Corollary 13) are proved in full in Appendix A. They are stated with their assumptions in Sections 4–5; the spectral sharpening of Proposition 17 is stated and proved in Appendix A with a compact summary in Section 5. Assumptions (e.g., non-negative edge weights with $W(G) > 0$, bounded distances $d(i,j) \in [0, D_{\max}]$, independent Bernoulli inclusion under Horvitz–Thompson weights) are stated inside the corresponding theorem statements and reused verbatim in the proofs.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: Section 6 summarizes the VMAS environment, team sizes, policy architecture, training algorithm (IPPO), checkpoints, rollout set size, and metric implementation. Appendix B lists every PPO hyperparameter and the $n{=}100$ scaling overrides. The frozen-init GPU timing sweep and the DiCo integration (Section 6.1, Appendix C, and Section 6.2) specify batch shape, timing regime (`torch.cuda.synchronize` on both sides of every `perf_counter`), number of trials, number of seeds, and configuration of the k-NN graph and the Bernoulli random graph. The $n{=}50$ head-to-head (Appendix D) reuses these hyperparameters except for the reduced per-iteration rollout (`on_policy_n_envs_per_worker`=120, `on_policy_collected_frames_per_batch`=12,000) and is driven by two matched launchers that share seeds, rollout shape, and set points: `ControllingBehavioralDiversity-fork/scripts/launch_n50_bern_setpoint_sweep.sh` for the Bernoulli-0.1 arm and `ControllingBehavioralDiversity-fork/scripts/launch_n50_full_setpoint_sweep.sh` for the full-SND arm; the head-to-head comparison, figure, and LaTeX table fragment are regenerated by `experiments/n50_bern_vs_full_comparison.py`. The post-hoc complete-graph SND audit in the same appendix is regenerated by `experiments/n50_posthoc_full_snd_validation.py` after the corresponding raw GPU logs are available. The non-VMAS transfer panel in Appendix C is reproduced by `experiments/mpe_ippo_training.py`, `experiments/mpe_measurement_panel.py`, and `scripts/launch_mpe_move2.sh`. Everything needed to reconstruct the code paths is in `graphsnd/metrics.py`, `graphs.py`, `training/train_navigation_batched.py`, and `experiments/exp2_timing_scaling.py` in the companion repository.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: The anonymous supplement contains the code, tests, curated result CSV/JSON/PDF artifacts, selected raw DiCo CSV logs, and a top-level `README.md` specifying installation, experiment reproduction commands, and a claim-to-code map. A preflight script (`scripts/preflight.sh`) runs the test suite and a smoke-training pass on each visible CUDA device to catch environment problems before overnight runs. For the anonymous submission the repository URL is anonymized in the main text; the public repository will be released upon acceptance.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: Training details (optimizer, learning rate, clip parameter, GAE λ, discount γ, rollout shape, mini-batch size, PPO epochs, hidden-layer width, activation) are listed in Appendix B. The DiCo integration of Section 6.2 additionally specifies the HetControl architecture, $\mathrm{SND}_{\mathrm{des}}$=0.1 set point, 167-iteration budget, three seeds, k=3 for the k-NN graph, Bernoulli inclusion probability p=0.1, and the 128-env subsample used for the k-NN construction. The n=50 head-to-head (Appendix D) uses the same 167-iteration budget, three seeds per cell, p=0.1 for the Bernoulli arm, and three set points $\mathrm{SND}_{\mathrm{des}} \in \{0.12, 0.14, 0.15\}$; the full-SND arm runs the same nine cells with the unmodified DiCo estimator on K_n (`model.diversity_estimator=full`) for a $2\times3\times3$=18-run paired comparison.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: The unbiasedness diagnostic (Section 6, Appendix C) reports 95% confidence intervals from 2,000 Bernoulli draws per p and the standardised deviation $|\mathrm{bias}|/\mathrm{SE}$ as a sharper test. The concentration diagnostic reports the empirical violation rate at δ=0.1 over 2,000 draws per cell and the 95th-percentile deviation. The timing experiments (Section 6.1, Appendix C) report mean, median, and standard deviation over 20 timing trials per cell after 3 warm-ups. The DiCo experiment (Section 6.2) reports mean $\pm$ standard deviation over three seeds for reward and applied SND, and median with IQR for per-call metric cost. The n=50 head-to-head (Appendix D, Table 7) reports late-window (last 50 iterations) means $\pm$ cross-seed standard error over three seeds per (set-point, estimator) cell and the relative tracking error $\overline{|\text{applied SND} - \mathrm{SND}_{\mathrm{des}}|}/\mathrm{SND}_{\mathrm{des}}$ separately for the Bernoulli-0.1 and full-SND arms; per-cell and aggregated summary CSVs are released alongside the LaTeX fragment. The post-hoc full-SND audit reports the same late-window statistics for the complete-graph audit signal in `results/dico_n50_posthoc_full_snd/`.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

Justification: The $n \in \{4, 8, 16\}$ frozen-checkpoint verifications (Section 6, Appendix C) run on a single CPU core with the reported 20 trials per cell in minutes. The $n{=}100$ overnight PPO run (Section 6.1) fits on one RTX 4090, as does the frozen-init $n \in \{100, 250, 500\}$ GPU timing sweep (Section 6.1, Appendix C). The DiCo closed-loop experiment (Section 6.2) also runs on one RTX 4090 at the reported 167-iteration budget. The $n{=}50$ head-to-head (Appendix D, 18 cells: three seeds × three set points × two estimators) runs on two RTX 4090 GPUs in parallel, at ≈2 GPU-hours per Bernoulli-0.1 cell (≈ 18 GPU-hours total) and ≈3.5 GPU-hours per full-SND cell (≈ 31.5 GPU-hours total, dominated by the full-SND aggregation over all $\binom{50}{2}{=}1{,}225$ pairs each forward). The post-hoc complete-graph audit repeats the same 18-cell grid with an additional logging callback, adding about 42 RTX 4090 GPU-hours. Including this audit, total compute for the experiments reported in this paper is under 150 GPU-hours on RTX 4090 class GPUs, plus a few CPU-hours for the $n \in \{4, 8, 16\}$ benchmarks and the MPE transfer panel (three CPU seeds at $n{=}10$ plus measurement passes).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: The work is theoretical and empirical research on a metric for behavioral diversity in cooperative MARL. It does not involve human subjects, sensitive data, or release of models with plausible misuse potential. No deviations from the NeurIPS Code of Ethics are required.

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    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [N/A]

    Justification: Graph-SND is a methodological metric contribution that reduces the wall-clock cost of measuring behavioral diversity in cooperative MARL. It enables neither new capabilities nor new deployments; it only makes an existing measurement cheaper. We therefore do not identify a direct path to positive or negative societal impact.

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: [N/A]

    Justification: No data or models with misuse potential are released; the released artifacts are experimental code, reproducible CSV/JSON result files, and paper figures.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: VMAS [Bettini et al., 2022] and DiCo [Bettini et al., 2024a] are credited in the main text and released under their original open-source licenses. The vendored fork of the DiCo codebase in `ControllingBehavioralDiversity-fork/` retains the upstream license, adds a `GRAPH_SND_CHANGES.md` file documenting the modifications, and includes the preferred citation in `CITATION.cff`. The original SND paper [Bettini et al., 2025] is cited throughout.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

Justification: The new assets are the `graphsnd` Python package, the two training scripts, the experiment scripts, and the released figure and CSV outputs. The top-level `README.md` documents installation, the four entry-point experiments, the overnight training launcher, and the repository layout, and cross-references the specific paper sections that each asset reproduces.

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